MATH 2400 - Multiple Integrals

This is the heart of it. This is what it's all about. How do you integrate things in higher dimensional spaces? That's the question. There are multiple answers.

15.1 Double Integrals over Rectangles.

The **double integral** of f over the rectangle R is

$$\iint_R f(x,y)\,dA = \lim_{m,n\to\infty} \sum_{i=1}^{m}\sum_{j=1}^{n} f(x_i^*, y_j^*)\Delta A$$

Students tend to not like Riemann Sums, which is a shame because they unify integration and give it geometric meaning. I think deep down it's a fear of the notation: limits and sums and those weird starred variables, oh, and also that delta thing. It's basically everything that Calculus students don't really understand all in one definition. No wonder there's such an almost instinctual aversion to them. But you've got to get over that. Each piece is a fairly simple geometric or arithmetic concept that taken together give an extremely elegant and intuitive meaning to what an integral is. Let's move from right to left.

- $\mathbf{\Delta A}$: This is the area of the tiny little subrectangle R_{ij}. That's all it is. Area, hence the A. Why the Δ? Because the way that you calculate area is to multiply two lengths together and the way that you calculate length is to find the *Difference* between two points. So $\Delta A = \Delta x \Delta y = (x_i - x_{i-1})(y_j - y_{j-1})$.
- $(\mathbf{x_i^*, y_j^*})$: This is an arbitrary point inside R_{ij}. Why arbitrary? Well, because the area of R_{ij} is shrinking to 0 so all points inside it are getting closer together. More will be said about this in a second, but basically, if f is "well-behaved" (which usually means continuous) then there can't be jumps between neighboring points. If the values of the independent variables are close, then the value of the dependent variable will be close. So we don't need to worry too much about the specific values involved, just that things are close. But because $\Delta A \to 0$, every point in R_{ij} will eventually be very, very close to every other point in R_{ij} and we don't have to worry about which point we select. Hence we can take an arbitrary point in R_{ij}, which we call (x_i^*, y_j^*).
- $\mathbf{f(x_i^*, y_j^*)}$: This is just height. That's all it is. The height of the surface given by $f(x,y)$ above the point (x_i^*, y_j^*) in the xy-plane. The function f is usually continuous, which means that there can't be too much difference in the heights at neighboring points. This extends to the idea that the heights at all points in a sufficiently small rectangle R_{ij} will be close, so we can just take any of them and get a pretty good approximation to all the others. Even if f is not continuous over all of R, it will be continuous over small enough portions of R that the same argument applies once the subrectangles R_{ij} are small enough that f is continuous over each one of them.
- $\mathbf{f(x_i^*, y_j^*)\,\Delta A}$: This is height times area and so gives volume. Volume of what? The rectangular solid lying above R_{ij}.
- $\mathbf{\sum_{i=1}^{m}\sum_{j=1}^{n}}$: This is adding together all of volumes of all of the little subrectangles R_{ij}. There are mn of them. Taken together, this gives an approximation to the volume under $f(x,y)$ over the rectangular region R.
- $\mathbf{\lim_{m,n\to\infty}}$: To move from an approximation to something exact you have to pass to the limit. Essentially what you're doing is letting the base of each subrectangle R_{ij} get smaller and smaller, but this is counterbalanced by the fact that there are now more of the R_{ij}. So even though each individual subrectangle is getting smaller, there are more of them in total. One thing is getting smaller (the size of each subrectangle) while simultaneously something else is getting bigger (the number of subrectangles). These contrasting changes perfectly balance each other and you get the exact value. This is the miracle

and beauty and power of limits. Two things, each which on its own would lead to something undesirable (an infinitely small rectangular solid having zero volume and the approximation of something curved by something flat), perfectly combine to give you exactly what you want.

Okay, that's enough of that. It might seem unnecessary to go into so much detail about the Riemann Sum, but until you come to terms with it, you won't really understand what an integral is and how they're all related to each other. So, please, take the time to really think about these things and try to make sense of them.

Now for some formulas:

- $\text{Area}(R) = \iint_R 1 \, dA$
- **Fubini's Theorem (aka Iterated Integrals):** If f is continuous on the rectangle $R = \{a \leq x \leq b, c \leq y \leq d\}$ then

$$\iint_R f(x,y) \, dA = \int_a^b \int_c^d f(x,y) \, dydx = \int_c^d \int_a^b f(x,y) \, dxdy$$

- If $f(x,y) = g(x)h(y)$ then

$$\iint_R g(x)h(y) \, dA = \left(\int_a^b g(x) \, dx \right) \left(\int_c^d h(y) \, dy \right)$$

15.2 Double Integrals over General Regions

This is a natural generalization of the definition for double integrals over rectangular regions, the only change being in the limits of integration.

If D is a **type 1 region** in the xy-plane then D can be written in the form $D = \{a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$ and we have that

$$\iint_D f(x,y) \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y) \, dydx$$

If D is a **type 2 region** in the xy-plane then D can be written in the form $R = \{c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$ and we have that

$$\iint_D f(x,y) \, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y) \, dxdy$$

The most important thing to notice here is that the limits of integration on the outer (or first) integral are **always** constants, while the limits of integration on the inner (or second) integral can be functions of the **outer** variable.

The most challenging part of integrating over a general region is changing the order of integration. Sketching the region D in the xy-plane is the best way to get a sense of what's going on. Depending on the order that you choose to integrate in, you'll need to invert the functions used to define the non-horizontal and non-vertical portions of the boundary of D. Just keep in mind that you need things on the inner integral to be defined in terms of the outer variable.

15.3 Double Integrals in Polar Coordinates

There are three important relations between x, y (rectangular coordinates) and r, θ (polar coordinates):

1. $x = r\cos\theta$
2. $y = r\sin\theta$
3. $x^2 + y^2 = r^2$

If f is continuous on a polar region of the form $R = \{\alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\}$ then

$$\iint_R f(x,y)\, dA = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta, r\sin\theta)\, r\, drd\theta$$

What this is saying is that when you change from rectangular to polar coordinates, the "area element" dA changes from $dxdy$ (or $dydx$) to $rdrd\theta$.

Intuitively, what this means is that the "size" of the polar rectangle scales by how far away it is from the origin. If you are very close to the origin, so that r is small, then the area of the polar rectangle is small. If you are far away from the origin, so that r is large, then the area of the polar rectangle is large. This makes complete sense if you draw it out.

15.4 Applications of Double Integrals

We don't really get very deep into this. But here are the formulas that you need to know.

- If $\rho(x,y)$ is the density at point (x,y) then the mass of a laminar region D is

$$m = \iint_D \rho(x,y)\, dA$$

- The moment about the x-axis is

$$M_x = \iint_D y\rho(x,y)\, dA$$

  The moment about the y-axis is

$$M_y = \iint_D x\rho(x,y)\, dA$$

- The coordinates $(\bar{x}, \bar{y})$ of the center of mass of a lamina occupying the region D are

$$\bar{x} = \frac{M_y}{m} = \frac{1}{m}\iint_D x\rho(x,y)\, dA$$

$$\bar{y} = \frac{M_x}{m} = \frac{1}{m}\iint_D y\rho(x,y)\, dA$$

- If $\sigma(x,y)$ is the charge density at a point (x,y), then the total electric charge, Q, in a region D is

$$Q = \iint_D \sigma(x,y)\, dA$$

15.5 Surface Area (also 16.6)

Think of this as the two-dimensional analogue of arc length. We will approximate the area of a surface by adding up the areas of a bunch of little parallelograms in the tangent planes. To calculate the area of the parallelograms, we need two vectors that give the sides of the parallelograms and then we take the norm of the cross product of these vectors.

If a smooth parametric surface S is given by the parameterization

$$\vec{r}(u,v) = \langle x(u,v), y(u,v), z(u,v)\rangle$$

with (u, v) in some domain D in the uv-plane, then the surface area of S is

$$A(S) = \iint_D |\vec{r}_u(u, v) \times \vec{r}_v(u, v)| \, dA$$

Please note that the double integral is taken over the uv-plane.

In the special case that the surface S is defined by $z = f(x, y)$ with (x, y) in D in the xy-plane, then the surface area is

$$A(S) = \iint_D \sqrt{[f_x(x, y)]^2 + [f_y(x, y)]^2 + 1} \, dA$$

Here, the integral is taken over the xy-plane.

15.6 Triple Integrals

This is a straight-forward generalization of double integrals, with all of the necessary complications that can occur when the dimension increases.

The triple integral over the rectangular solid $B = \{a \leq x \leq b, c \leq y \leq d, r \leq z \leq s\}$ is defined by

$$\iiint_B f(x, y, z) \, dV = \lim_{l,m,n \to \infty} \sum_{i=1}^{l} \sum_{j=1}^{m} \sum_{k=1}^{n} f(x_i^*, y_j^*, z_k^*) \Delta V$$

All of the terms in the Riemann Sum are the exact analogue of the terms in the Riemann Sum for the double integral.

Just as for double integrals, there is a version of Fubini's Theorem (aka Iterated Integrals) for triple integrals:

$$\iiint_B f(x, y, z) \, dV = \int_a^b \int_c^d \int_r^s f(x, y, z) \, dzdydx$$

Please note that there are now $3! = 6$ ways to order the variables: $dzdydx, dydzdx, dzdxdy, dxdzdy, dxdydz, dydxdz$. There's a lot of flexibility here. The main thing to keep in mind is that you have to have the limits of integration match the order in which you are integrating.

Things become more complicated when integrating over a non-rectangular solid region E. The outer integral must still have constant limits of integration, but both of the other integrals can have variables in their limits; the middle integral can have the outer variable in its limits, while the inner integral can have both of the other variables in its limits. For example, if we are integrating in the order $dzdydx$ then you can have

$$\iiint_E f(x, y, z) \, dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{h_1(x,y)}^{h_2(x,y)} f(x, y, z) \, dzdydx$$

There are type 1, 2, and 3 solid regions. These correspond to projecting down to the coordinate planes:

- type 1 solid regions have the form $E = \{(x, y) \text{ in } D_1, h_1(x, y) \leq z \leq h_2(x, y)\}$. Here you are projecting onto the xy-plane and D_1 can be either a type 1 or type 2 plane region (in the xy-plane).
- type 2 solid regions have the form $E = \{(y, z) \text{ in } D_2, h_1(y, z) \leq x \leq h_2(y, z)\}$. Here you are projecting onto the yz-plane and D_2 can be either a type 1 or type 2 plane region (in the yz-plane).
- type 3 solid regions have the form $E = \{(x, z) \text{ in } D_3, h_1(x, z) \leq y \leq h_2(x, z)\}$. Here you are projecting onto the xz-plane and D_3 can be either a type 1 or type 2 plane region (in the xz-plane).

Changing the order of integration for triple integrals is much more challenging than for double integrals for two reasons:

1. There are now 6 different orders to consider.
2. It is much harder to visualize the solid region E.

Sometimes you can visualize the solid region by finding the three projections onto the coordinate planes, but even if you have these, it can be hard to "put them together" in your mind's eye.

Usually you'll want to work with the inequalities used in the description of E as a set of points and try to rearrange the inequalities, thinking of the maximum and minimum values that the variables take as you move through the solid E. This takes practice and the best thing to do is to work through examples until you start to develop a "feel" for it.

An important geometric application of triple integrals is finding the volume of the solid region E:

$$V(E) = \iiint_E 1 \, dV$$

15.7 Triple Integrals in Cylindrical Coordinates

Cylindrical coordinates are a combination of polar (in two variables) and rectangular (in the remaining variable). The usual ordering is

$$x = r\cos\theta, \; y = r\sin\theta, \; z = z$$

but you can have any of the variables be the "rectangular" variable. For example, if E is circular in the yz-plane, then you would have

$$x = x, \; y = r\cos\theta, \; z = r\sin\theta$$

Converting from integrals in rectangular to cylindrical coordinates goes through the same as for converting from integrals in rectangular to polar coordinates:

$$\iiint_E f(x,y,z) \, dV = \int_\alpha^\beta \int_{g_1(\theta)}^{g_2(\theta)} \int_{h_1(r\cos\theta, r\sin\theta)}^{h_2(r\cos\theta, r\sin\theta)} f(r\cos\theta, r\sin\theta, z) \; r \; dz dr d\theta$$

You will almost always have θ as the outer variable.

The important thing to note here is the change of volume element dV from $dzdydx$ to $r \; dzdrd\theta$. The r is doing the same thing here that it did in polar: scaling the "cylindrical rectangles" according to how far away they are from the rectangular axis.

15.8 Triple Integrals in Spherical Coordinates

The things to keep in mind with spherical coordinates are the following:

$$x = \rho\cos\theta\sin\phi, \; y = \rho\sin\theta\sin\phi, \; z = \rho\cos\phi$$

and the very useful

$$x^2 + y^2 + z^2 = \rho^2$$

When changing from an integral in rectangular to spherical coordinates, you have

$$\iiint_E f(x,y,z) \, dV = \int_\alpha^\beta \int_{g_1(\theta)}^{g_2(\theta)} \int_{h_1(\theta,\phi)}^{h_2(\theta,\phi)} f(\rho\cos\theta\sin\phi, \rho\sin\theta\sin\phi, \rho\cos\phi)\rho^2\sin\phi \; d\rho d\phi d\theta$$

You will almost always have ρ as the inner variable and θ as the outer variable (thereby forcing ϕ to be the middle variable).

The important thing to note here is that the volume element dV changes from $dzdydx$ to $\rho^2 \sin\phi \; d\rho d\phi d\theta$. Geometrically, the ρ^2 term is scaling the "spherical rectangle" by it's distance-squared from the origin, and the $\sin\phi$ term is scaling according to how far you've rotated down from the north pole of the sphere. Again, the best thing to do is draw some pictures to "see" this.

15.9 Change of Variables in Multiple Integrals

The key here is the Jacobian. For a double integral, we have $(x(u,v), y(u,v))$ and the Jacobian is given by

$$\frac{\partial(x,y)}{\partial(u,v)} = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}$$

Then we have that

$$\iint_R f(x,y)\, dA = \iint_S f(x(u,v), y(u,v)) \left|\frac{\partial(x,y)}{\partial(u,v)}\right| dudv$$

The region S is in the uv-plane and the region R is in the xy-plane.

For a triple integral, we have $(x(u,v,w), y(u,v,w), z(u,v,w))$ and the Jacobian is given by

$$\frac{\partial(x,y,z)}{\partial(u,v,w)} = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{pmatrix} = \text{something too horrible to type up}$$

Then we have that

$$\iiint_R f(x,y,z)\, dV = \iiint_S f(x(u,v,w), y(u,v,w), z(u,v,w)) \left|\frac{\partial(x,y,z)}{\partial(u,v,w)}\right| dudvdw$$

The solid region S is in uvw-space and the solid region R is in xyz-space.

If you think of changing coordinates as a change of variables (which is what it is) then you will find that the Jacobian that corresponds to the change from rectangular to polar is r, the Jacobian that corresponds to the change from rectangular to cylindrical is also r, and the Jacobian that corresponds to the change from rectangular to spherical is $\rho^2 \sin\phi$.

- General Things to Keep In Mind
  - With any multiple integral, you probably want to start by thinking about what coordinate system to use. If you decide on spherical, then the order of integration is usually fixed: $d\rho d\phi d\theta$, though sometimes the order of the two angular variables can change. If you use cylindrical or polar, you will almost always integrate r before θ.
  - If you are integrating over a region that has any kind of spherical boundaries, then spherical is probably the way to go. If you see $x^2 + y^2 + z^2$ anywhere in the integrand, then you'll probably also want to use spherical.
  - Same kind of thing goes for cylindrical, though what you're looking for now is anything of the form $a^2 + b^2$ where a, b are any two of x, y, z. The distinction here is that you're only seeing sums of squares of two of the three variables.
  - If you're sticking with rectangular coordinates, then you really need to think about the order of integration. This will usually be determined by the shape of the region being integrated over (is it type 1, 2, or 3) and the function being integrated.
  - Don't forget that changing coordinates is really changing variables and you need to include the Jacobian. Just have them memorized for polar (r), cylindrical (r), and spherical ($\rho^2 \sin\phi$).